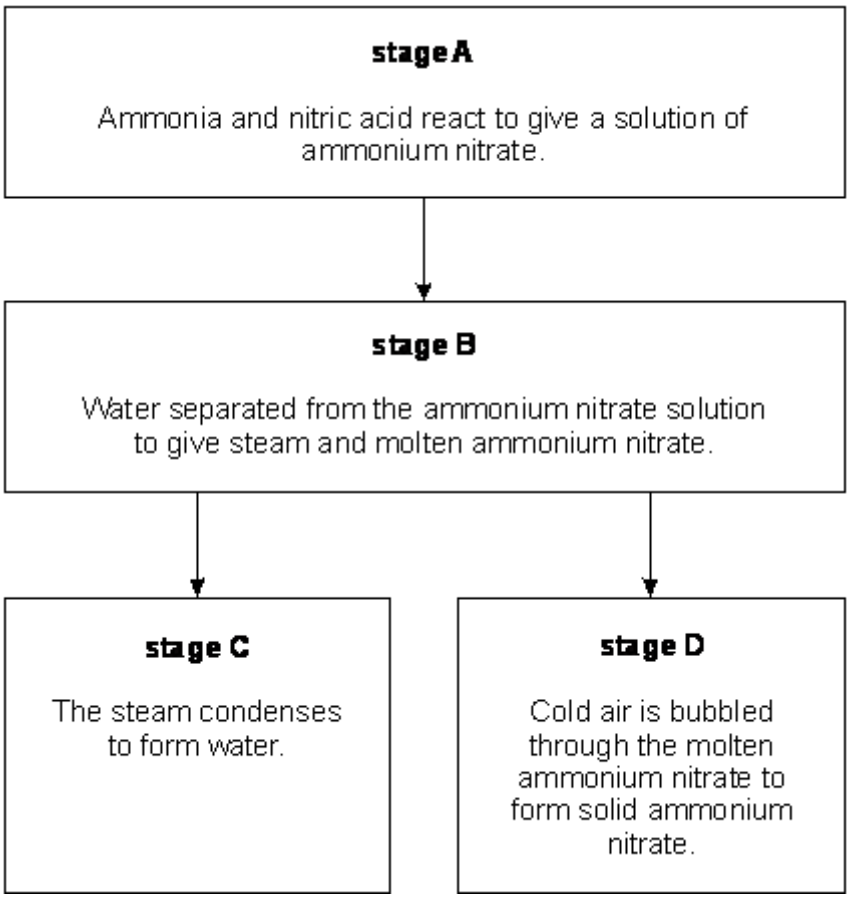
Q1. Ammonium nitrate is used as a garden fertiliser. It is manufactured by the reaction between ammonia gas and nitric acid. The diagram below represents stages in the process for making ammonium nitrate.



(a) Ammonia is an alkaline gas. In stage A, an alkali reacts with an acid. What effect does an alkali have on an acid?

.....

1 mark

(b) The formula for ammonium nitrate is NH_4NO_3.

(i) How many different elements are there in ammonium nitrate?

.....

1 mark

(ii) How many atoms are represented in the formula for ammonium nitrate?

.....

1 mark

(c) Potassium nitrate is also a fertiliser. It can be made from the reaction between potassium hydroxide and an acid.
Complete the word equation for the reaction, naming the acid used.

potassium hydroxide + ... → potassium nitrate + ...

2 marks
Maximum 5 marks

Q1.

(a) it neutralises it

accept 'neutralisation'

accept 'it produces heat'

1 (L6)

(b) (i) 3

1 (L7)

(ii) 9

1 (L7)

(c) nitric acid → water

answers must be in the correct order

2 (L7)

[5]